HW 6 Due 11/15

① 1) $t = x - 80$ $\beta = 50$ $\alpha = 2$

mean $= \alpha\beta = 100 \Rightarrow 100 + 80 = 180$

variance $= \alpha\beta^2 = 2 \cdot 2500 = 5000$

2) $P(X < 200)$ or $P(X' < 120)$ $\int v\,dw = vw - \int w\,dv$

$\int_{80}^{200} f(x)\,dx = \int_{80}^{200} \frac{x-80}{50^2} e^{\frac{-x-80}{50}}\,dx$ $u = \frac{x-80}{50}$ $du = \frac{dx}{50}$

$x = 200$, $u = 12/5$; $x = 80$, $u = 0$

$\int_0^{12/5} \frac{50u}{50^2} e^{-u} \cdot 50 \cdot du = \int_0^{145} ue^{-u}\,du = (u)(-e^{-u}) - \int -e^{-u}\,du$ (v, dw)

$= [ue^{-u}]_0^{12/5} + [-e^{-u}]_0^{12/5}$

$\int e^{-u}\,du = -e^{-u}$ (w)

$= -\frac{12}{5}e^{-12/5} + \left(-e^{-12/5} + 1\right)$

$= -0.22 + 0.909 = 0.69$ 69%

② 1) $\mu = 6$ $6^2 = 25$ $6 = 5$ $\frac{x-6}{5}$

$x_1 = 0$ $x_2 = 8/5 = 1.6$ $P(z \le 1.6) = 0.9452$

$P(z \le 0) = 0.5$

0.4452

2) $x_1 = -2/5 = -0.4$ $x_2 = 1.6$ $P(z \le -0.4) = 1 - P(z \le 0.4) = 0.34458$

$0.9452 - 0.34458 = 0.60062$

3) $x_1 = -2$ $x_2 = -6/5 = -1.2 \rightarrow 1 - 0.88493 = 0.11507$

$1 - 0.9772 = 0.0228$ $0.11507 - 0.0228 = 0.09227$

4) $9/5 = 1.8$ $1 - 0.96407 = 0.03593$

5) $P(1 < X < 11)$ $x_1 = -1$ $x_2 = 1 \rightarrow 0.84134$

0.15866

$0.84134 - 0.15866 = 0.68268$

6) $P(-4 < X < 16)$ $x_1 = -2$ $x_2 = 2$ $0.97725 - 0.02275 = 0.9545$

7) $P(-9 < x < 21)$ $x_1 = -3$ $x_2 = 3$ $0.99865 - 0.00135 = 0.9973$

8) $P(-6.4 < x < 18.4)$ $x_1 = -2.48$ $x_2 = 2.48$ $0.99379 - 0.00621 = 0.98758$

≈ 2.50 ≈ 2.50

$$-\frac{1}{2\sigma^2}(x-\mu)^2 \xrightarrow{\text{derivative}} -\frac{1}{\sigma^2}(x-\mu)$$

③ pdf $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$

$$f'(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot \exp\left[\frac{-(x-\mu)^2}{2\sigma^2}\right] \cdot \left[\frac{-(x-\mu)}{\sigma^2}\right]$$

$$= \underbrace{\frac{-(x-\mu)}{\sigma^3\sqrt{2\pi}}}_{u} \cdot \underbrace{\exp\left[\frac{-(x-\mu)^2}{2\sigma^2}\right]}_{v}$$

$f''(x) = u'v + uv'$

$$= \left(\frac{-1}{\sigma^3\sqrt{2\pi}}\right)\exp\left[\frac{-(x-\mu)^2}{2\sigma^2}\right] + \left(\frac{-(x-\mu)}{\sigma^3\sqrt{2\pi}}\right)\left(\frac{-(x-\mu)}{\sigma^2}\right)\exp\left[\frac{-(x-\mu)^2}{2\sigma^2}\right]$$

$$= \left(\frac{-1}{\sigma^3\sqrt{2\pi}}\right)\exp\left[\frac{-(x-\mu)^2}{2\sigma^2}\right] + \frac{(x-\mu)^2}{\sigma^5\sqrt{2\pi}}\exp\left[\frac{-(x-\mu)^2}{2\sigma^2}\right]$$

$$= \underbrace{\left[\exp\left[\frac{-(x-\mu)^2}{2\sigma^2}\right]\cdot\frac{1}{\sigma^5\sqrt{2\pi}}\right]}_{\text{can't be 0}}\left((x-\mu)^2-\sigma^2\right)$$

$(x-\mu)^2 - \sigma^2 = 0$

$(x-\mu)^2 = \sigma^2$

$x-\mu = \pm\sigma$

$x = \mu+\sigma,\ \mu-\sigma$ □ ✓

④ $\mu = 21.37$ $\sigma^2 = 0.16$ $\sigma = 0.4$ $\quad z = \frac{X-21.37}{0.4}$

1) $z = \frac{22.07-21.37}{0.4} = 1.75$

$P(z > 1.75) = 1 - P(z \le 1.75) = 0.04006$

2) $\frac{20.857-21.37}{0.4} \approx -1.28$ $\quad P(z \le -1.28) = 0.10027$

$Y \sim Bin(15, 0.10027)$ $\quad P(Y \le 2) = P(Y=0) + P(Y=1) + P(Y=2)$

$$\binom{15}{0}(0.10027)^0(1-0.10027)^{15} + \binom{15}{1}(0.10027)^1(1-0.10027)^{14} + \binom{15}{2}(0.10027)^2(1-0.10027)^{13}$$

$0.204967 + 0.342636 + 0.267294 = 0.814897$

⑤ 1) $P(X=x) = \frac{1}{4}$ $P(Y=y) = \frac{1}{4}$

$f(x,y) = \frac{1}{4} \cdot \frac{1}{4} = \boxed{\frac{1}{16}}$

$\forall x \in S_X, y \in S_Y$

$y: \sum_{x=1}^{4} \frac{1}{16} = \boxed{\frac{1}{4}} = f_Y(y), \quad y \in \{1,2,3,4\}$

x: 2) $f_X(x) = \sum_{y \in S_Y} f(x,y) =$

$\sum_{y=1}^{4} \frac{1}{16} = \boxed{\frac{1}{4}}$

$x \in \{1,2,3,4\}$

3) $P(X=x) \cdot P(Y=y) \stackrel{?}{=} P(X=x, Y=y)$

$f_X(x) \cdot f_Y(y) = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16} = p(x,y)$

$\frac{1}{16} = \frac{1}{16}$ ✓

Yes, they are independent since both sides are equal.

⑥ 1) $f_X(x) = P(X=x) = \sum_{y=1}^{4} \frac{x+y}{32} = \frac{x+1+x+2+x+3+x+4}{32} = \boxed{\frac{4x+10}{32}}$

$f_X(1) = \frac{14}{32} = \frac{7}{16}$

$f_Y(2) = \frac{13}{32} = \frac{9}{16}$

2) $f_Y(y) = P(Y=y)$

$= \sum_{x=1}^{2} \frac{x+y}{32} = \frac{1+y+2+y}{32} = \boxed{\frac{3+2y}{32}}$

$f_X(x) = \begin{cases} \frac{7}{16}, & x=1 \\ \frac{9}{16}, & x=2 \end{cases}$

$f_Y(1) = \frac{5}{32}$ $f_Y(2) = \frac{7}{32}$ $f_Y(3) = \frac{9}{32}$ $f_Y(4) = \frac{11}{32}$

$f_Y(y) = \begin{cases} \frac{5}{32}, & y=1 \\ \frac{7}{32}, & y=2 \\ \frac{9}{32}, & y=3 \\ \frac{11}{32}, & y=4 \end{cases}$

3) $f(1,1) = \frac{1}{16}$ $f_X(1) \cdot f_Y(1) = \frac{7}{16} \cdot \frac{5}{32} = \frac{35}{512}$

$\frac{1}{16} \neq \frac{35}{512}$, so X and Y are not independent, since $P(x,y) \neq f_X(x) f_Y(y)$ for $x=1, y=1$.